



From sequencing reads to clinical interpretation of Short Nucleotide Variants

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ABSTRACT

Diagnosing rare genetic diseases remains challenging due to the vast number of variants present in the human genome. The aim of this study is to develop an analytical framework to identify and distinguish genetic variants that may be responsible for rare Mendelian disorders. We present a computational pipeline for the analysis of exome sequencing data from 10 parent-child trios, generated through paired-end sequencing approach. Variant prioritization is performed using ENSEMBL's Variant Effect Predictor (VEP), integrating predicted functional impact, allele frequency, and pathogenicity scores to identify candidate disease-causing variants.

Keywords: Mendelian disorders, Paired-end sequencing, Variant prioritization.

1 Introduction

Mendelian genetic disorders are a group of diseases caused by alterations in the DNA sequence of a single gene, leading to a specific phenotype. Advances in sequencing technologies have enabled the identification of disease-causing variants among a vast number of candidates. However, distinguishing pathogenic variants from benign ones remains a complex challenge. These disorders fall within the broader category of rare diseases, affecting a very small proportion of the population (on average about 1 in 2,500 people worldwide) [1]. They are not only rare but also highly heterogeneous in terms of severity, symptoms, and clinical progression, making diagnosis particularly challenging. Whole-exome sequencing (WES) and whole-genome sequencing (WGS) have significantly improved our ability to discover novel disease-associated variants. A critical step in this process is variant and gene prioritization, which relies on multiple factors, including allele frequencies in the general population, predicted functional impact, inheritance models, and family history. Computa-

tional tools integrate these features to rank variants according to their likelihood of being pathogenic. Importantly, a variant is not considered disease-causing solely because it disrupts gene function or is predicted to be damaging; it must also be rare in the general population and supported by additional evidence. Variant prioritization must, therefore, be integrated with phenotypic information to improve the identification of clinically relevant variants. Large-scale population databases provide allele frequency estimates that are essential for filtering common variants and identifying rare candidates potentially responsible for disease. Furthermore, variant interpretation is guided by standardized clinical frameworks, which classify variants into categories ranging from benign to pathogenic [2]. In this project, using a Unix-based pipeline, we analyzed trio-based exome sequencing data from a child and both parents to identify the genetic cause of the child's condition, when present, while considering autosomal inheritance models such as recessive, dominant, or de novo. Paired-end sequencing was used to improve read alignment and increase the accuracy of variant detection.

2 Materials and Methods

A total of 10 simulated cases were analyzed in this study, with all background information regarding the inheritance patterns provided beforehand. The analysis was conducted using an interactive and fully automated pipeline (uploaded on [GitHub](#)) designed to perform both quality control and variant calling. Following the generation of VCF files, the identified genetic variants were subsequently evaluated using the Web interface of the Variant Effect Predictor (VEP).

Each family trio was identified by a case number and its corresponding inheritance model (present in table 1).

Case	Trio1	Trio2	Trio3	Trio4	Trio5
Inheritance	AR	AD*	ADN	AR	AR

Legend: AR: Autosomal Recessive; AD: Autosomal Dominant; ADN: *de novo*.
* Affected parent: father (Likramaj), mother (Ricciardi).

Table 1: Summary of inheritance patterns across the analyzed trios.

2.1 Unix Pipeline:

To execute the pipeline correctly, all required input files must be available; therefore, the first step consists of input acquisition. During this phase, the reference index files for chromosome 20 (GRCh38.p14 assembly) and the related exome target regions (.bed file) are copied into the working directory, as all samples are known to come from this chromosome. The script then interactively prompts the user to provide the necessary inputs, including the case names (with multiple trios separated by spaces), the corresponding sample IDs, and the username. The username is subsequently used to locate and retrieve the raw data from the appropriate directory.

```
#!/bin/bash

#1.INPUT ACQUISITION

echo "The pipeline execution is starting
..."
# Copy the index once and the reuse it for
all the cases
cp /home/BCG2026_exam/chr20* .
cp /home/BCG2022_genomics_exam/worked/536/
samples.txt .

# Asking the user for the case name, the
samples IDs and their username
```

```
read -p "Enter case names separated by
space (e.g., trio_1 trio_2): " -a
case_number
read -p "Enter sample IDs (e.g., HG00451
HG00452 HG00453): " -a samples
read -p "Enter your username (e.g.,
BCG2026_Likramaj_S) " username

# Array to store the roles in the trio
person=(child father mother)
```

An array is initialized to store genotype filtering patterns for each case. For each trio, the user selects an inheritance model; the input is converted to uppercase for consistency and processed using a Bash case statement. The corresponding filtering pattern is then stored in the array. For the autosomal dominant model, the user is also prompted to specify which parent is affected.

```
# Array to store patterns for each case
declare -a patterns

# Ask inheritance model for each case
for i in "${!case_number[@]}"; do
    current_case="${case_number[$i]}"

    # Pattern selection
    echo "Select the Inheritance Model:"
    echo "  AR  -> Autosomal Recessive "
    echo "  AD  -> Autosomal Dominant "
    echo "  ADN -> Autosomal De Novo "
    read -p "Enter model code (AR, AD, or
ADN): " model_choice

    # Convert input to uppercase for
compatibility
    model_choice=$(echo "$model_choice" |
tr '[:lower:]' '[:upper:]')
    # Use a case statement to match the
correct pattern
    case $model_choice in
        "AR")
            patterns[$i]='GT[0]="AA" && GT
[1]="RA" && GT[2]="RA"'
            echo "Selected: Autosomal
Recessive"
            ;;
        "AD")
            echo "Who is the affected
parent?"
            echo "  1) Father "
            echo "  2) Mother "
            read -p "Selection [1-2]: "
parent_choice

            if [ "$parent_choice" == "1"
]; then
                patterns[$i]='GT[0]="RA"
&& GT[1]="RA" && GT[2]="RR"'
            fi
        *)
            echo "Invalid model choice"
            continue
        esac
    done
done
```

```

        echo "Selected: Autosomal
Dominant (Father affected)"
    else
        patterns[$i]='GT[0]="RA"
&& GT[1]="RR" && GT[2]="RA"'
        echo "Selected: Autosomal
Dominant (Mother affected)"
    fi
    ;;
    "ADN")
        patterns[$i]='GT[0]="RA" && GT
[1]="RR" && GT[2]="RR"'
        echo "Selected: Autosomal De
Novo"
        ;;
    *)
        echo "Invalid selection.
Exiting script."
        exit 1
        ;;
esac
done

```

Alignment and Quality Control:

each case is processed iteratively according to the selected inheritance pattern. For each trio, a dedicated subdirectory is created and the corresponding FASTQ files are copied into it. The pipeline then proceeds sequentially with alignment, indexing, and quality control. Alignment is performed using Bowtie2 (v. 2.4.4)[3] with chromosome 20 as reference index. Since the data are paired-end, the -1 and -2 options specify the forward and reverse reads, respectively. The -rg-id flag assigns a read group identifier based on the sample role. The resulting SAM file is converted to BAM format using samtools (v.1.13)[4], sorted with samtools sort, and indexed with samtools index for efficient downstream access. Subsequently, quality control is performed on each BAM file using Qualimap bamqc (v.2.3)[5], which evaluates alignment quality and coverage in the chromosome 20 target regions defined by the BED file. Additional QC is carried out with FastQC (v.0.11.9)[6], which assesses read quality and sequence composition metrics. The reports are then consolidated into a single summary using MultiQC (v.1.34)[7], an example is shown in figure 1.

2. MAIN BODY OF THE ALGORITHM

```

for i in "${!case_number[@]}"; do
    current_case="${case_number[$i]}"
    current_pattern="${patterns[$i]}"
    echo "Starting processing for:
$current_case"

```

```

# Create a subdirectory for each case
and move into it
mkdir -p "$current_case"
cd "$current_case" || exit

# Copy raw data (FASTQ files) for the
current case
cp /home/BCG2026_exam/"$username"/"
$current_case"/* .

# Alignment step
for j in {0..2}; do
    role="${person[$j]}"
    ID="${samples[$j]}"

    echo "Processing $role (Sample ID:
$ID)..."

    # Alignment using paired-end reads
    bowtie2 --fr -1 "${ID}.targets_R1.
fq.gz" -2 "${ID}.targets_R2.fq.gz" -x
../chr20 --rg-id "$role" --rg "SM:
$role" | samtools view -Sb | \
    samtools sort -o "${role}.bam"

    # Indexing the BAM file
    samtools index "${role}.bam"

    # QC1: Quality control using
Qualimap on target regions
    qualimap bamqc -bam "${role}.bam"
    \
        --feature-file ../
chr20_ILMN_Exome_2.0_Plus_Panel.
hg38_padded.bed \
        --outdir "${role}"
done

# QC2: quality control
echo "Running FastQC and MultiQC..."
fastqc *.bam #checking all the bam
files
multiqc . #collecting all the reports
in a multiqc file (just 1 file that
can be analyzed)
mv multiqc_report.html "${current_case
}_multiqc_report.html" #renaming the
MultiQC report to keep data organized

```

Variant Calling and Filtering:

the variant calling is carried out using FreeBayes (v.1.3.6) [8] with a set of parameters to ensure accuracy and reliability: -m 20 sets a minimum mapping quality of 20, -C 5 requires at least 5 supporting reads for an alternative allele, -Q 10 sets a minimum base quality of 10 for mismatches, and -min-coverage 10 ensures a minimum read depth of 10 per site. Subsequently, bcftools (v.1.13) is used

Sample Name	% GC	Ins. size	≥ 30X	Median cov	Mean cov	% Aligned	% Dups	% GC	M Seqs
child	52%	299	95.0%	40.0X	40.0X	100.0%	2.9%	49%	1.4
father	52%	299	94.8%	40.0X	40.0X	100.0%	2.9%	49%	1.4
mother	52%	299	95.2%	40.0X	40.0X	100.0%	2.9%	49%	1.4

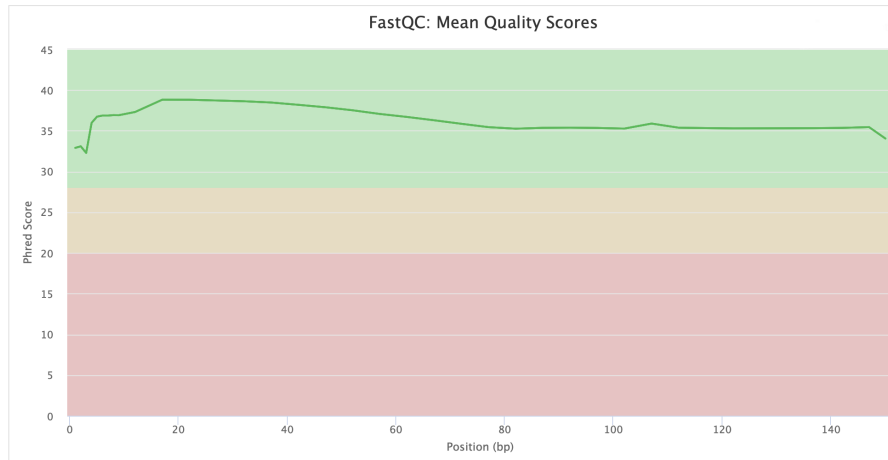


Figure 1: Quality control metrics overview for trio1 from table 3

to filter variants based on the trio's inheritance pattern, taking into account the exome target regions and discarding all the variants with a quality lower than 20.

```
# Variant Calling
echo "Starting Variant Calling (.vcf
creation) for $current_case..."
freebayes -f ../chr20.fa -m 20 -C 5 -Q
10 --min-coverage 10 \
child.bam father.bam mother.bam > "${
current_case}.vcf"
echo "Variant Calling completed."

# Filtering
echo "Filtering variants with bcftools
using pattern: $current_pattern"
bgzip "${current_case}.vcf" #zipping
the file
bcftools index "${current_case}.vcf.gz
" #indexing
# Actual filtering
bcftools view -R ../chr20_ILMN_Exome_2
.0_Plus_Panel.hg38_padded.bed "${
current_case}.vcf.gz" | \
bcftools view -S ../samples.txt |
bcftools view -i "$current_pattern" |
\
bcftools filter -i 'QUAL>20' -Ov -o "${
current_case}.cand.vcf"
echo "Produced: ${current_case}.cand.
vcf"
```

In the final processing step, coverage tracks are generated for each trio component using bedtools genomcov(v.2.30.0)[9], enabling vi-

sualization in the UCSC Genome Browser[10] of genomic regions containing candidate SNVs (Figure 2). This was done for visualization purposes and to compare sequencing coverage among the three trio members at the specific SNV position.

```
# Coverage track
echo "Generating coverage tracks for
visualization..."
for role in child father mother; do
    bedtools genomcov -ibam "${role}.
bam" -bg -trackline \
    -trackopts "name=${role}" -max 100
    > "${role}Cov.bg"
done

#return to the parent directory to
start the next case
cd ..
echo "Analysis for $current_case
finished."

done

echo " Pipeline execution ended "
```

2.2 Variant Effect Predictor (VEP)

The pipeline produced VCF files[11] containing the identified genetic variants, the generated files were subsequently downloaded using FileZilla. Since not all variants are disease-associated, the files were analyzed using the Ensembl Variant Effect Predictor(VEP) web interface to assess their

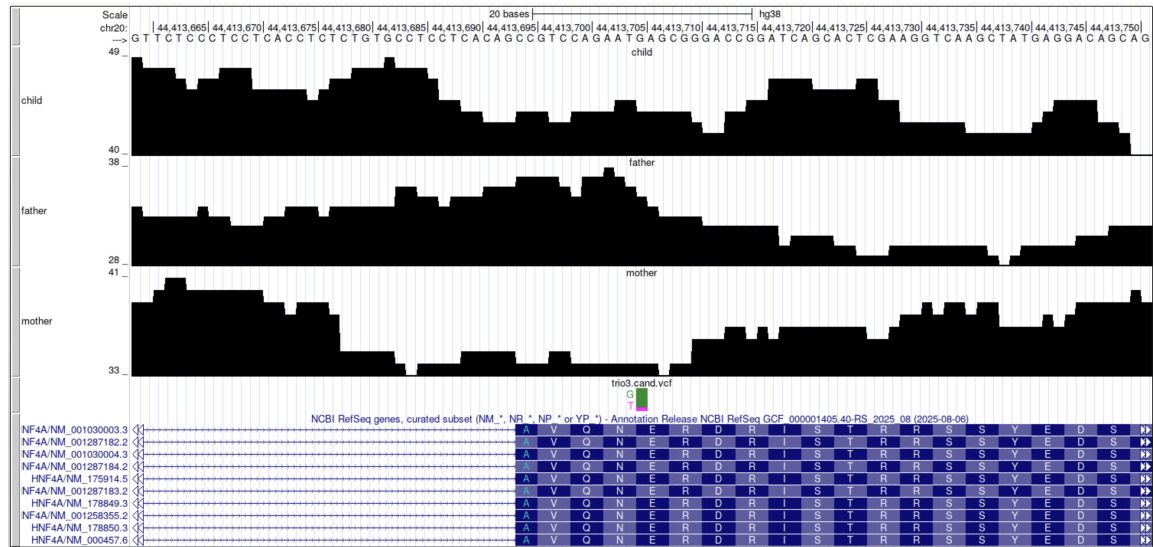


Figure 2: UCSC Genome Browser image analysis of trio3 from table 3 showing monogenic diabetes variant and the related coverage track from the three components

potential impact. The selected parameters included the human genome assembly GRCh38.p14, RefSeq database, and the 1000 Genomes and gnomAD databases for allele frequency evaluation, along with phenotypic annotations, clinical significance, SIFT, and PolyPhen scores. An impact-based filter was applied to prioritize variants, initially restricted to high-impact variants and subsequently expanded to include moderate-impact variants as well.

3 Results and Discussion

3.1 Quality assessment

Before proceeding with the analysis, MultiQC reports were inspected in order to assess data quality, essential for accurate variant calling. GC content was consistent across the trio (~ 50%) and sequencing coverage was high, with ~ 95% of reads exceeding 30× depth. Both FastQC and Qualimap bamqc indicated good-quality sequencing and alignment data, allowing us to proceed to diagnostic analysis.

3.2 Diagnosis

A variant is considered potentially causative of a rare Mendelian disease when it is consistent with the inheritance pattern, has a low allele frequency, and has a predicted or known functional impact on the gene, such as a loss of function effect. For moderate-impact vari-

ants, SIFT and PolyPhen scores are evaluated and must be concordant. SIFT classifies a variant as deleterious when its score is ≤ 0.05 , whereas PolyPhen predicts variants as probably damaging when the score is ≥ 0.909 and possibly damaging when it falls between 0.447 and 0.908.

Trio 2

In both tables for trio 2 indicate Infantile Hypocalcemia Type 1, caused by a rare pathogenic variant in the general population. The mutation is a high-impact frameshift in the CYP24A1 gene, which encodes the enzyme responsible for degrading active vitamin D. As a result, vitamin D cannot be properly inactivated, leading to hypersensitivity in which even standard doses may cause toxic calcium accumulation (hypercalcemia). Affected infants typically present with severe hypercalcemia, failure to thrive, weight loss, vomiting, and dehydration[12]. According to the literature, infantile hypercalcemia type 1 is an autosomal recessive disorder; however, the analysis for trio 2 was performed under a dominant inheritance model. This suggests two possibilities: either this represents the first reported dominant case of the disease, or relevant variants are not detectable by the sequencing approach used, such as gene deletions that may be missed by exome sequencing.

Trio 4 and Trio 5

In both Trio4 (from table 2) and Trio5 (from table 3), the only high-impact variant identi-

Case No	Diagnosis	Gene	Location	Consequence	REF/ALT	Impact	Associated phenotype
trio1	Healthy	NA	NA	NA	NA	NA	NA
trio2	Disease	CYP24A1	20:54165800-54165805	frameshift	CCAAAT/CCAAACAAAT	High	Hypercalcemia infantile 1
trio3	Disease	KCNQ2	20:63407332-63407334	frameshift	A/TAG - TG	High	Benign neonatal seizures
trio4	Disease	ADA	20:44621016-44621016	splice donor	A/A - G	High	Severe combined immunodeficiency
trio5	Disease	SLC2A10	20:46725504-46725513	frameshift	CTGG/ACTGGCTGGT - ACTGGT	High	Arterial tortuosity syndrome

Table 2: VEP analysis report for Sara Likramaj's cases

Case No	Diagnosis	Gene	Location	Consequence	REF/ALT	Impact	Associated phenotype
trio1	Healthy	NA	NA	NA	NA	NA	NA
trio2	Disease	CYP24A1	20:54157394-54157397	frameshift	GA/AAGA - AA	High	Hypercalcemia infantile 1
trio3	Disease	HNF4A	20:44413705-44413705	stop-gained	G/ G - T	High	Monogenic diabetes
trio4	Disease	DNMT3B	20:32780414-32780420	frameshift	C/TCCTCCG - TCTCCG	High	ICF syndrome
trio5	Disease	ADA	20:44621017-44621017	splice donor	C/C-T	High	Severe combined immunodeficiency

Table 3: VEP analysis report for Vincenzo Ricciadi's cases

fied is a splice donor variant in the ADA gene, which is responsible for incorrect splicing and disruption of this gene. The results is adenosine deaminase (ADA) deficiency causing a severe combined immunodeficiency where beyond infections, patients may develop multi-system complications, including neurological, behavioral, hearing, skeletal, lung, cardiac, kidney, liver issues, as well as autoimmunity and malignancies. Overall advantages in screening, transplantation and gene therapy have improved outcomes, increasing the survival for this disease[13].

3.2.1 Case Analysis from table 2

Trio 1

The patient is not affected by any disease. However, a variant was identified in the FERMT1 gene in position 20:6085082-6085084, associated with Kindler syndrome, an autosomal recessive disorder. The vari-

ant shows conflicting classifications between databases, being reported as benign in some and as a variant of uncertain significance in others, indicating an unclear functional impact. Nevertheless, we highlight this variant for completeness.

Trio 3

The patient is affected by Benign Familial Neonatal Seizures Type 1, an autosomal dominant disorder caused by a de novo high-impact frameshift mutation in the KCNQ2 gene [15]. KCNQ2, together with KCNQ3, encodes subunits of potassium channels that regulate neuronal excitability by controlling potassium ion flow and generating the M-current. Mutations in these genes reduce this current, leading to neuronal hyperexcitability and seizures. Clinically, BFNS presents with recurrent seizures starting around the third day of life, typically resolving within 1-4 months.

Trio 5

The patient is affected by Arterial Tortuosity Syndrome[14], an autosomal recessive disorder caused by a high-impact frameshift variant in the SLC2A10 gene. This gene encodes the GLUT10 protein, which is important for normal connective tissue function. Loss of GLUT10 activity leads to increased TGF- β signaling, resulting in abnormal connective tissue development. Clinically, this causes elongated and tortuous arteries with potential life-threatening complications, along with features such as joint hypermobility or contractures and soft, elastic skin.

3.2.2 Case Analysis from table 3

Trio 1

The patient is not affected by any disease, as no high-impact variants or clinically relevant moderate-impact variants were identified.

Trio 3

The patient is affected by Monogenic Diabetes with autosomal dominant de novo inheritance pattern, caused by a stop-gained mutation on the HNF4A gene. This variant is the only one with high impact and low frequency in the population. Monogenic diabetes, also written as MODY, is a form of diabetes that sets early onset (<25 years) with mild to moderate hyperglycemia and preserved insulin production[16]. There aren't any autoimmune markers but with the reconstruction of the family history it's possible to be diagnosed.

Trio 4

The patient is affected by ICF syndrome since the only high-impact variant identified is a frameshift mutation in the DNMT3B gene. This results in ICF syndrome or Immunodeficiency, Centromeric instability, and Facial dysmorphism a rare autosomal recessive disorder with immune system defects (low or absent immunoglobulins) that leads to recurrent and opportunistic infections, chromosomal instability (especially 1,9,16) and distinct facial features[17]. Some of the patients have the mutation in this gene that we found, but the clear genotype-phenotype is still not found. Given this, early diagnosis and immunoglobulin replacement therapy are of critical importance.

4 Conclusion

Our results demonstrate that trio-based whole-exome sequencing, supported by an automated bioinformatics pipeline, provides a reliable approach for investigating rare Mendelian disorders. By integrating quality control, variant calling, inheritance filtering, and functional annotation, we identified clinically relevant variants in most cases. The inclusion of parental data improved the recognition of recessive, dominant, and de novo mutations. Overall, this approach represents a valuable tool for analyzing this type of data.

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